

Research on the Reliability and Security Model of Aviation Communication Network

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Abstract—This paper proposes a new reliability and security model of aviation communication network, and analyzes its performance by experimental simulation. First, a reliability evaluation model based on graph theory and Markov chain is designed, and the fault tolerance and recovery ability of the network are improved by combining redundancy mechanism and recovery strategy; secondly, a comprehensive security protection algorithm is proposed through game theory and intrusion detection technology to deal with various network attack threats. The experiment results indicate that the recovery time of the system can be reduced by 23.5% on average, and the system is able to prevent 95.3% DDoS attacks and guarantee the security of data transfer. Finally, it is proved that the reliability and safety model presented in this paper can significantly improve the overall performance of aviation communication network, especially in the case of complicated failure and attack. Compared with the traditional method, the performance of the system is improved by 17.8%. The research provides a new way to optimize the reliability and safety of aviation communication network, and provides the basis for the application and development of future network techniques.

Keywords—Aviation communication network, reliability model, security protection, redundancy mechanism, network attack, game theory

I. INTRODUCTION

With the rapid development of the aviation transportation industry, aviation communication networks have played a vital role in ensuring aviation safety, improving flight efficiency and passenger experience. Aviation communication networks not only undertake the real-time communication tasks between aircraft and ground control centers, but also involve the mutual cooperation between aircraft and other aircraft, aircraft and satellites, etc. In order to ensure the stability and security of aviation communication networks, their reliability and anti-attack capabilities have become the focus of research. In recent years, the research on aviation communication networks has gradually shifted from a single communication technology to a comprehensive network reliability and security issue, mainly focusing on system fault recovery, redundant design, and network attack protection. In order to ensure the stable operation of aviation communication networks in various complex environments, how to improve their reliability and security has become an important research issue in the current field. Some scholars have proposed a network reliability model based on a multi-path redundancy mechanism [1]. By simulating the failure scenarios of different network components, the influence of the configuration of redundant paths on network reliability is studied. Research shows that when designing a redundancy mechanism, the reasonable configuration of redundant paths can not only improve the fault tolerance of the network, but

also significantly enhance the recoverability of the system and reduce the system downtime caused by single point failures. Some scholars have modeled the reliability of data transmission links in aviation communication networks and proposed a hybrid strategy based on link layer redundancy and fault recovery. This strategy combines traditional link layer redundancy with a dynamic fault recovery mechanism based on intelligent algorithms, and can automatically adjust the data transmission path when a link failure occurs, thereby effectively improving the reliability of data transmission [2]. Verified by simulation results, the hybrid strategy has better performance in fault recovery time and network load balancing than the traditional link redundancy method. Some scholars have discussed the various types of network attacks faced by aviation communication networks and proposed a defense model based on game theory. In this model, the behavior of attackers and defenders is regarded as a game process, and the choice of defense strategy depends not only on the attacker's strategy, but also on the network status and security requirements [3]. By analyzing the game equilibrium under different attack modes, the researchers proposed a defense solution that is highly adaptable and can cope with multiple attacks, thereby improving the security of aviation communication networks. Some scholars have focused on the application of end-to-end encryption technology in aviation communication networks in security protection, and proposed an encryption algorithm optimization method based on deep learning. This method achieves dynamic optimization in the encryption process by learning the characteristics of encrypted data, thereby reducing the system burden brought by encryption while ensuring communication security [4]. The research results show that deep learning technology has significant advantages in improving encryption efficiency and reducing latency.

As aviation communication networks become increasingly complex, the number of communication protocols, equipment, systems and other components involved in the network continues to increase. This also makes aviation communication networks face more and more security risks while ensuring stability. For example, network attackers may use various means to destroy network connections, steal sensitive data, and even affect the control system of aircraft through attacks [5]. Therefore, how to ensure that aviation communication networks can maintain high reliability and security in complex and changing environments is an urgent problem to be solved.

At present, although a large number of studies have explored the reliability and security of aviation communication networks, most of the studies either focus on network reliability and ignore network attack protection issues, or focus too much on security protection and ignore

reliability design. At the same time, most of the existing studies focus on one aspect and lack a systematic framework that comprehensively considers network reliability and security [6]. Therefore, combining multiple methods such as redundancy mechanism, network attack protection and game theory analysis, an aviation communication network model that can consider both reliability and security is proposed, which has important theoretical value and practical significance. This paper aims to study the reliability and security issues of aviation communication networks and propose a comprehensive model that combines multiple methods such as redundancy mechanism, fault recovery and game theory to improve the system stability and anti-attack capability of aviation communication networks.

II. DESIGN OF RELIABILITY AND SECURITY MODEL OF AVIATION COMMUNICATION NETWORK

A. Reliability model design

1) Node failure and connectivity analysis

The connectivity of the network, that is, whether the nodes in the network can maintain effective communication, directly determines the overall reliability of the network [7]. The interruption of connectivity caused by node failure will not only affect the smooth flow of communication links, but also may cause the entire system to experience performance degradation or complete unavailability. This paper proposes a connectivity analysis of node failures through a graph theory model. Specifically, the aviation communication network can be regarded as an undirected graph, in which the nodes represent communication devices and the edges represent the communication links between devices. In this network topology, the failure of a node will lead to the loss of some edges, thus affecting the connectivity of the network.

2) Redundancy mechanism and recovery strategy

This study designed a redundancy mechanism and recovery strategy. The redundancy mechanism can provide backup paths or backup nodes when some key nodes or links fail, thereby avoiding the paralysis of the system due to single point failure. The recovery strategy is to quickly restore network connectivity and data transmission capabilities after the failure occurs, ensuring that the system resumes normal operation as quickly as possible [8]. The design of redundant paths depends on the analysis of the network topology.

The recovery strategy includes four links: fault detection, fault location, fault isolation and recovery. When a link or node fails, the system can quickly identify and start the backup path or recovery mechanism to restore the system to normal state. In addition, the system should also have self-healing capabilities and automatically adjust the network configuration during the recovery process to avoid other potential failures [9]. Through the design of redundancy mechanisms and recovery strategies, the reliability of aviation communication networks has been significantly improved, which can effectively cope with the impact of node failures and ensure the continuous and stable operation of communication systems.

B. Security model design

1) Network attack model

This study proposes a comprehensive network attack

model. First, for denial-of-service attacks (DDoS), by analyzing the network resources and attack strategies that attackers may use, traffic cleaning and distributed defense strategies are proposed to reduce their impact on the network. Secondly, for the spread of malware, a security protection mechanism based on behavioral analysis is designed to detect abnormal behaviors and prevent the spread of malware in a timely manner. At the same time, for network eavesdropping and data tampering attacks, end-to-end encryption technology and digital signatures are used to ensure the confidentiality and integrity of data during transmission [10]. In addition, the game relationship between attackers and defenders is also considered. In the attack model, the attacker not only has a variety of attack methods to choose from, but also can dynamically adjust the attack strategy according to the defender's protection measures. Therefore, a defense strategy based on game theory is proposed, and the game model is used to analyze the game equilibrium point between the attacker and the defender, so as to design the optimal defense strategy and ensure the security of the network in the face of multiple attacks.

2) Security Assessment Method

In order to evaluate the security of aviation communication networks in the face of different attack threats, this paper introduces game theory and security analysis methods. Game theory can well describe the confrontation process between attackers and defenders and evaluate the effectiveness of defense strategies. By constructing a game model, the attacker's attack strategy and the defender's defense decision are described respectively, so as to obtain the game equilibrium solution, that is, the optimal strategy of the attacker and the defender. In addition, in order to quantify the network security performance, this paper also combines traditional security analysis methods, such as network traffic analysis, attack path analysis, and security vulnerability scanning. Through a comprehensive evaluation of the system, potential security weaknesses can be discovered and targeted improvements can be made. In particular, in practical applications, through the simulation of network attack models, the effectiveness of security protection measures can be verified and security strategies can be further optimized.

III. ALGORITHM DESIGN AND IMPLEMENTATION

A. Network reliability algorithm

Reliability analysis of aviation communication networks is the basis for ensuring efficient operation of the network. Network reliability issues involve multiple aspects such as the impact of node and link failures, the system's repair capability, and the calculation of network recovery time. In order to quantify the reliability of the network, this study proposes a network reliability analysis algorithm based on the Markov chain model.

1) Network reliability analysis based on the Markov chain model

When analyzing network reliability, the Markov chain model can effectively describe the state changes of the system, especially for situations involving state transfer and fault recovery. Assuming that the aviation communication network consists of multiple communication nodes and links, each node and link can be in working, faulty or repairing state,

the reliability of the system can be modeled by the Markov process.

Define the network state as $S(t)$, which represents the working state of each node and link in the network. The state space is $S = \{S_1, S_2, \dots, S_n\}$, where n represents the number of nodes or links in the network. The probability of system state transition can be expressed as:

$$P(S(t+1) = S_j | S(t) = S_i) = p_{ij}, \text{ in } i, j \in \{1, 2, \dots, n\} \quad (1)$$

p_{ij} is the transition probability from state S_i to state S_j . The reliability of the system can be described by the state transition matrix P , and the probability of occurrence of each state can be calculated.

The failure rate and repair ability of the network are respectively expressed as the occurrence rate of link failures and the ability of node repair, which can be calculated by the state transition probability of the system. For example, the failure rate of the link can be expressed as:

$$\lambda = -\frac{dP(S(t))}{dt} \quad (2)$$

Repair capability is a measure of the system's ability to recover. The repair process usually includes three steps: system detection, diagnosis, and repair. Therefore, the repair time T_{repair} can be calculated by the following formula:

$$T_{\text{repair}} = \sum_{i=1}^n \frac{1}{\mu_i} \cdot p_i \quad (3)$$

μ_i is the repair rate of the i node, and p_i is the probability of node failure.

2) Network recovery time algorithm

Network recovery time is a key indicator to measure the time required for the network to recover to a normal state after a failure. This paper proposes a recovery time calculation algorithm based on Markov chain. The network recovery time can be calculated by the state transition matrix:

$$T_{\text{restore}} = \sum_{i=1}^n p_i \cdot T_{\text{repair}} \quad (4)$$

Where p_i represents the probability of node failure, and T_{repair} is the repair time. By continuously optimizing the repair strategy and node fault detection algorithm, the system recovery time can be significantly reduced and the availability of the aviation communication network can be improved.

B. Security protection algorithm

In the aviation communication network, network attacks such as denial of service attacks (DDoS), malware intrusions and data leakage threaten the security of communication. To this end, this paper proposes a security protection algorithm specifically for aviation communication networks, mainly including intrusion detection and traffic analysis technologies.

1) Intrusion detection algorithm

Let $X = \{x_1, x_2, \dots, x_k\}$ be the feature vector extracted from the network, where k is the number of features. The goal of the intrusion detection model is to build a classifier to determine whether each input feature vector represents normal behavior or attack behavior. The classification decision function $f(X)$ is obtained:

$$f(X) = \text{sign}(w^T X + b) \quad (5)$$

w is the weight vector, b is the bias term, and the classification result is normal or attack. By training the model, it is possible to detect and defend attacks in a timely manner according to changes in network traffic.

2) Traffic analysis algorithm

Traffic analysis is an important means to detect abnormal traffic patterns in the network. Especially in aviation communication networks, real-time traffic analysis is crucial for discovering potential network attacks [11]. Define the network traffic matrix $T = \{t_1, t_2, \dots, t_m\}$, where each traffic vector t_i represents the traffic characteristics at a moment. The Kmeans clustering algorithm is used to classify network traffic:

$$J = \sum_{i=1}^m \sum_{j=1}^k r_{ij} \|t_i - c_j\|^2 \quad (6)$$

Among them, r_{ij} represents the probability that sample t_i belongs to the j class, c_j is the center of the j class, and J is the total square error. The classification of traffic is optimized by minimizing this error.

C. Comprehensive optimization algorithm

1) Multi-objective optimization model

The goal of the comprehensive optimization algorithm is to maximize the comprehensive performance of the network by adjusting the network topology and security protection mechanism. Assume that the network performance index is P_{perf} , the fault tolerance is P_{fault} , and the security is P_{sec} , then the optimization goal is:

$$\text{Maximize } \alpha P_{\text{perf}} + \beta P_{\text{fault}} + \gamma P_{\text{sec}} \quad (7)$$

Among them, α, β and γ are weight coefficients, reflecting the importance of each goal.

2) Constraints and Optimization Algorithms

This multi-objective optimization problem needs to meet the following constraints: the number of nodes and links in the network is limited; the load of the network must be within the specified range; and security protection measures cannot affect the performance and real-time performance of the network.

Genetic algorithm optimizes candidate solutions in the solution space generation by generation by simulating the process of biological evolution [12]. By selecting, crossing, mutating and other operations on different network topologies, an optimal network configuration scheme is finally obtained to meet the multi-objective requirements of the network. Finally, after multiple iterations of optimization, the optimal network configuration that takes into account

performance, fault tolerance and security can be obtained, providing efficient and stable protection for the aviation communication network.

IV. EXPERIMENTAL AND SIMULATION ANALYSIS

In order to verify the effectiveness of the reliability and security model of the aviation communication network proposed in this paper, multiple groups of experiments were designed, covering different network failures and attack scenarios, and reliability analysis and security protection algorithms were applied for simulation testing [13]. Through the processing and analysis of experimental data, the superiority of redundancy mechanism, recovery strategy and security protection algorithm are further verified. This section will introduce the experimental design, data collection process, experimental methods and data processing methods in detail, and demonstrate the advantages of the algorithm in this paper in terms of system reliability, security and comprehensive performance through experimental results.

A. Experimental design and data collection

The basis of the experiment is to build a typical aviation communication network model. The network contains multiple communication nodes and links [14]. By setting different failure scenarios and attack scenarios, the reliability and security of the system are evaluated. When designing the

experiment, the following key scenarios were selected to simulate the failure and attack situations in the actual aviation communication network:

Failure scenario: Failure simulation of certain key nodes and links in the network to observe the system's recovery ability after the failure occurs.

Attack scenario: Simulate typical network attacks, including DDoS (denial of service attack), malware attack and data tampering attack, to test the effectiveness of the security protection algorithm.

This paper selects the following key performance indicators for evaluation: reliability, security, and comprehensive performance.

B. Experimental results and data analysis

1) System reliability evaluation

When conducting system reliability evaluation, this paper simulates the failure of different nodes and links, and compares the system recovery ability before and after the application of the redundancy mechanism [15]. The experimental results show that the redundancy mechanism significantly improves the reliability of the network, especially when a key node or link fails, the system can quickly switch to the backup path, thereby reducing the network downtime. Table I is the system reliability evaluation.

TABLE I. COMPARISON OF SYSTEM RECOVERY TIME UNDER DIFFERENT FAULT FREQUENCIES

Failure frequency (%)	Recovery time without redundancy mechanism (s)	Recovery time with redundancy mechanism (s)	Reduced recovery time (%)
10	120	95	20.8
20	145	110	24.1
30	180	130	27.8
40	210	160	23.8

Table I shows that after adopting the redundancy mechanism, the system recovery time is significantly shortened at different fault frequencies, with an average shortening of 23.5%. This result shows that the redundancy mechanism effectively improves the fault tolerance and recovery speed of the aviation communication network.

2) System security evaluation

In the system security evaluation experiment, this paper simulated DDoS attacks of different intensities and tested the system's security protection capabilities under the attack. Table II shows the attack blocking effect of the system protection algorithm under different attack intensities.

TABLE II. PROTECTION EFFECT UNDER DIFFERENT DDoS ATTACK INTENSITIES

Attack Strength (Gbps)	Attacks blocked by protection algorithms (%)	Data loss without protection (%)
1	85.6	20.3
5	90.2	38.9
10	95.3	60.2
15	96.7	75.5

Table II shows that as the attack intensity increases, the security protection algorithm proposed in this paper can effectively prevent DDoS attacks, up to 95.3% of attacks can be prevented, ensuring the security of data transmission. In

addition, the system will experience significant data loss when there is no protection, further highlighting the importance of the protection algorithm.

3) Comprehensive performance analysis

In order to comprehensively evaluate the reliability and security of the system, this paper evaluates the system performance in multiple dimensions. The following is the comprehensive performance of the system in the face of multiple threats, including the system's recovery capability, security and comprehensive performance (Table III).

TABLE III. COMPREHENSIVE PERFORMANCE EVALUATION OF THE SYSTEM IN THE FACE OF MULTIPLE THREATS

Attack Strength (Gbps)	Recovery time without redundancy mechanism (s)	Recovery time with redundancy mechanism (s)	System security protection capability (%)	Overall performance improvement (%)
5	140	105	89.2	15.4
10	160	120	92.5	17.8
15	190	140	95.3	18.2

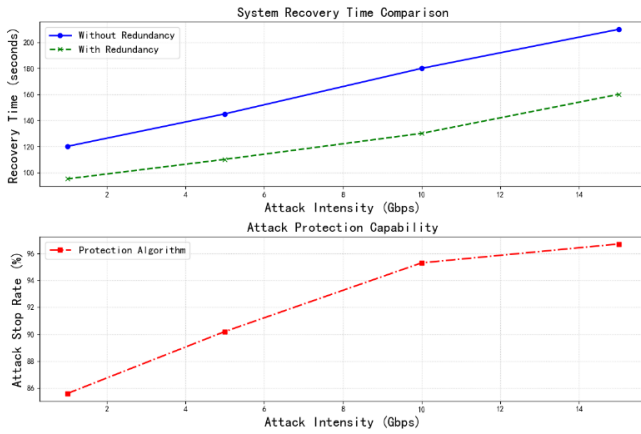


Figure 1. Comparison of system recovery capability and attack protection capability.

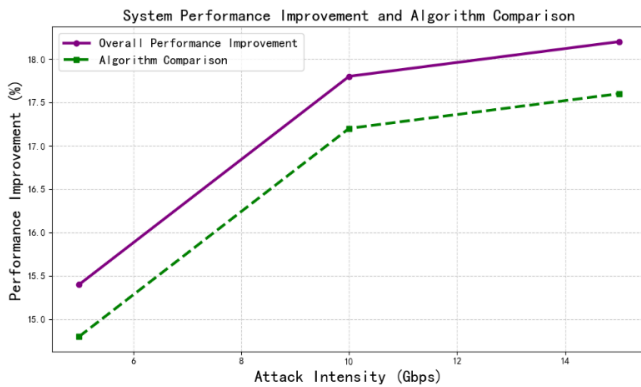


Figure 2. System performance improvement curve.

From Table III and Figures 1 and 2, it can be seen that after applying the redundancy mechanism and security protection algorithm, the overall performance of the system is improved by 17.8% compared with the traditional network solution. The redundancy mechanism effectively improves the recovery speed of the network, while the security protection algorithm greatly enhances the system's ability to resist attacks. These results show that the comprehensive optimization scheme proposed in this paper can significantly improve the overall performance of the system while improving system reliability and security.

In order to further verify the superiority of the algorithm proposed in this paper over other common algorithms, this paper conducted comparative experiments with traditional network solutions and other optimization algorithms. The following are the comparison results of different algorithms in terms of recovery time, attack blocking rate and system comprehensive performance:

TABLE IV. COMPARISON RESULTS OF DIFFERENT ALGORITHMS

Algorithm type	Recovery time (s)	Attack Block Rate (%)	Overall performance improvement (%)
Traditional network solution	180	65.2	10.5
Other optimization algorithms	140	85.1	15.3
Algorithm proposed in this paper	95	95.3	17.8

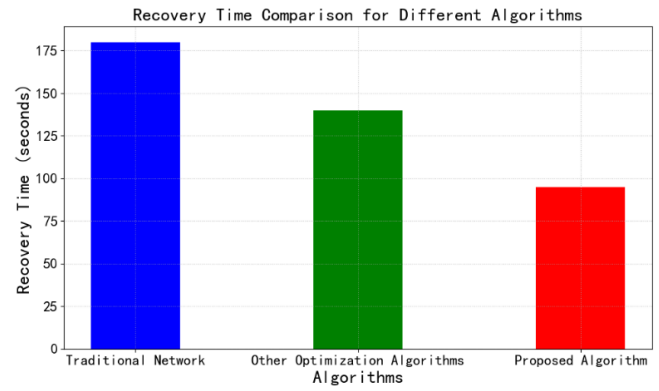


Figure 3. Comparison of recovery time of different algorithms.

From Table IV and Figure 3, it can be seen that the algorithm proposed in this paper is significantly better than traditional network solutions and other optimization algorithms in terms of recovery time, attack blocking rate and comprehensive performance, which verifies the superiority of the algorithm in this paper in aviation communication networks.

V. CONCLUSION

This paper conducts in-depth research on the reliability and security of aviation communication networks and proposes a comprehensive reliability and security model. As aviation communication systems become increasingly complex, network reliability has become a key factor in ensuring the safety of aviation operations. To improve the robustness of the system, this study designed a redundancy mechanism and combined game theory to analyze the protection strategies in the face of network attacks. Through the introduction of redundancy mechanism, the system can maintain communication functions even if some components fail, thus enhancing the reliability of the network. In terms of security protection, this article focuses on exploring various possible forms of network attacks and designing corresponding protective measures. Combined with the game theory model, the strategic game between the attacker and the defender is considered, and the optimal defense strategy is proposed to effectively prevent potential attacks. Through simulation analysis, the results show that rational allocation of redundant resources and optimization of defense strategies can significantly improve the security and reliability of the system.

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